Mostafa Bamdad

Researcher in Computational Mechanics | PhD Candidate

Bauhaus University Weimar, Germany

Bruckner-Str. 15, 99423 Weimar, Germany

Email | +49 176 64389666 | GitHub | Website | LinkedIn

Profile

Researcher and PhD Candidate in Computational Mechanics with expertise in scientific machine learning, neural operators, physics-informed modeling, and PDE-based simulation. Experienced in developing reliable computational models for nonlinear, multiphysics, and time-dependent systems using Python, PyTorch, JAX, MATLAB, Abaqus, and FEniCS. Research profile combines numerical methods, mechanics, spectral representations, simulation workflows, and engineering interpretation, with applications in structural mechanics, phase-field modeling, and scientific machine learning.

Education

Since 06/2022	PhD Candidate in Computational Mechanics, Bauhaus University Weimar, Germany Supervisor: Prof. Timon Rabczuk | h-index: 6 | Citations: 252
09/2016–06/2019	M.Sc. Mechanical Engineering, University of Kashan, Iran
09/2012–09/2016	B.Sc. Mechanical Engineering, University of Guilan, Iran

Core Expertise

- Scientific machine learning for PDE-based and mechanics-driven systems
- Physics-informed neural networks and neural operators, including PENCO, PINOs, VINO, FNOs, F-FNOs, DeepONets, and custom architectures such as HAMNO
- Surrogate modeling for nonlinear PDE systems and long-horizon spatiotemporal simulation
- Spectral analysis and feature extraction using Fourier-based methods
- Numerical analysis, multiphysics modeling, and computational mechanics
- FEM/FEA-based simulation workflows, mesh-based simulation, numerical validation, error analysis, and scientific visualization

Technical Skills

Programming	Python, NumPy, SciPy, Matplotlib, PyTorch, TensorFlow, JAX, MATLAB
ML/DL & SciML	PINNs, FNOs, operator learning, hybrid physics-data methods, neural operators
Simulation Workflows	FEM/FEA, mesh-based simulation, Abaqus, FEniCS
Modeling & Analysis	PDE modeling, numerical validation, sensitivity/error analysis, scientific visualization
VC/OS	Git / Linux, Windows
Languages	English: fluent | German: advanced, B2/C1 | Persian: native

Selected Research Projects

- **HAMNO** – Developed a hierarchical adaptive multi-scale neural operator for nonlinear dynamical systems with multi-scale structures, long-range interactions, and stable long-horizon prediction. The framework combines local and global representations to improve simulation accuracy and physical consistency in time-dependent PDE systems.
- **PENCO** – Developed a physics–energy–numerics-consistent operator for stable and accurate 3D phase-field simulation. The framework improves long-term prediction, data efficiency, and physical reliability in complex spatiotemporal PDE systems. Implementation available at github.com/MBamdad/PENCO.
- **Scientific Machine Learning for Elastic Mechanics** – Developed JAX-based scientific machine learning models for PDE-based structural systems, connecting physics-based modeling, numerical methods, and data-driven simulation for engineering applications. Code available at DeepNetBeam.

Awards and Training

- DAAD Scholarship, German Academic Exchange Service, 2023–Present
- Python for Computational Mechanics
- ABAQUS CAE – Advanced Static and Dynamic Simulation

Selected Publications

- **Developer of PENCO**: Physics–Energy–Numerics-Consistent Operator for 3D phase-field modeling, *CMAME*, 2026.
  doi.org/10.1016/j.cma.2026.118862
- Scientific Machine Learning for PDE-Based Structural Systems, *Neurocomputing*, 2024.
  doi.org/10.1016/j.neucom.2024.129119
- Multiphysics Modeling of Magneto-Electro-Elastic Sandwich Structures, *Advances in Nano Research*, 2023.
  doi.org/10.12989/anr.2023.15.5.423
- Multiphysics vibration analysis of SMA-hybrid sandwich beams on elastic foundations.
  doi.org/10.1007/s00707-020-02697-5
- Analysis of temperature-dependent material: magneto-electro-elastic beam.
  doi.org/10.1177/1077546319860314
- Numerical and multiphysics modeling of electro-elastic response of piezoelectric composite pressure vessels.
  doi.org/10.1016/j.compstruct.2019.111119
- Analysis of carbon nanotube reinforced composite beams and experimental tensile testing of nanocomposites.
  doi.org/10.12989/scs.2018.29.3.405

Reference

Prof. Timon Rabczuk
Chair of Computational Mechanics, Institute of Structural Mechanics
Bauhaus-Universität Weimar
Email: timon.rabczuk@uni-weimar.de

Mostafa Bamdad
Senior ML Researcher
Bauhaus University Weimar
Computational Mechanics Department
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